Stein's Identity and the Unbiased Estimator of Risk

STAT 220B – Lecture 9

Where we left off

Lecture 8 established that for $\bar{X} \sim \mathcal{N}(\theta, \sigma^2/n)$ with $\theta \in \mathbb{R}$, the sample mean is **minimax** under squared-error loss. The proof used Theorem 18 with a sequence of $\mathcal{N}(0, \tau^2)$ priors.

That argument extends without change to $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ in p dimensions: the sample mean $\delta^0(\mathbf{x}) = \mathbf{x}$ has constant risk p (sum of p coordinate variances) and is minimax for **every** p.

But Example 46 (Lecture 5) flagged a deeper issue: for $p \geq 3$, this minimax estimator is **inadmissible**. James and Stein (1961) wrote down an estimator that uniformly dominates the sample mean.

Question for today: How does one *prove* that a candidate estimator dominates the sample mean? Stein invented the right machinery in 1973–1981. Today we develop it.

Reading: Berger, Section 5.4.1 (motivation) and Section 5.4.2 (the unbiased estimator of risk).

The Stein phenomenon (Berger $§$5.4.1)

The basic setup

Let $\mathbf{X} = (X_1, \ldots, X_p)^T \sim \mathcal{N}_p(\theta, I_p)$, where $\theta = (\theta_1, \ldots, \theta_p)^T \in \mathbb{R}^p$ is unknown.

Estimate θ under **sum-of-squares loss**:

$$L(\theta, \mathbf{a}) \;=\; \|\theta - \mathbf{a}\|^2 \;=\; \sum_{i=1}^{p} (\theta_i - a_i)^2.$$

The standard estimator. $\delta^0(\mathbf{x}) = \mathbf{x}$. This is:

- The MLE
- The UMVUE
- The posterior mean under any sensible noninformative prior
- The minimax estimator (Lecture 8 argument, p-dimensional version)

Its risk is constant:

$$R(\theta, \delta^0) \;=\; \mathbb{E}_\theta \|\mathbf{X} - \theta\|^2 \;=\; \sum_{i=1}^{p} \mathrm{Var}_\theta(X_i) \;=\; p.$$

The surprise (Stein 1956, James-Stein 1961)

For $p = 1$ or 2: δ^0 is admissible (proof later in the course).

For $p \geq 3$: there exists a rule δ^{JS} with $R(\theta, \delta^{JS}) < p$ for all $\theta \in \mathbb{R}^p$.

Specifically, the **James-Stein estimator**:

$$\delta^{JS}(\mathbf{x}) \;=\; \left(1 - \frac{p-2}{\|\mathbf{x}\|^2}\right)\mathbf{x}.$$

Every coordinate is shrunk toward zero by the common factor $1 - (p-2)/\|\mathbf{x}\|^2$. We will show next lecture that

$$R(\theta, \delta^{JS}) \;=\; p - (p-2)^2 \mathbb{E}_\theta\left[\frac{1}{\|\mathbf{X}\|^2}\right] \;<\; p.$$

For now, focus on the conceptual question: why would this work, and how would one ever find it?

Why this seems impossible

The Stein phenomenon broke the classical paradigm. Three apparent puzzles.

Puzzle 1: The coordinates look independent. Estimating θ_1 from X_1, θ_2 from X_2, etc., feels like p separate one-dimensional problems. Each is solved by X_i separately (which is admissible in 1D). Why should combining them allow improvement?

Puzzle 2: δ^{JS} uses irrelevant information. To estimate θ_1, $\delta^{JS}(\mathbf{x})_1 = (1 - (p-2)/\|\mathbf{x}\|^2)x_1$ depends on $x_2, x_3, \ldots, x_p$ through $\|\mathbf{x}\|^2$. Why should data about unrelated parameters help?

Puzzle 3: It works for any θ. Bayesian-style shrinkage toward a presumed center only beats the MLE when θ is actually near that center. But δ^{JS} dominates δ^0 at **every** $\theta \in \mathbb{R}^p$, including θ very far from the origin.

The resolution is geometric: in dimension ≥ 3, the volume of $\mathcal{N}_p(\theta, I_p)$ concentrates on a thin shell at radius $\sqrt{p}$ from θ. Shrinking that shell slightly toward any fixed point produces an estimator closer to θ in expected squared distance.

Strategy for the proof

To prove $R(\theta, \delta^{JS}) < R(\theta, \delta^0)$ for all θ, we need to compute risk for both estimators. The standard tool would be:

$$R(\theta, \delta) \;=\; \int_{\mathbb{R}^p} \|\delta(\mathbf{x}) - \theta\|^2 \, \phi_p(\mathbf{x} - \theta) \, d\mathbf{x},$$

where ϕ_p is the $\mathcal{N}_p(0, I_p)$ density. For $\delta = \delta^{JS}$, the integrand contains terms like $1/\|\mathbf{x}\|^2$, products of $\mathbf{x}$ and $1/\|\mathbf{x}\|^2$, and so on. Direct integration is unpleasant.

Stein's idea (1973, 1981): rewrite the risk *without* the integral over θ. Find a function $\hat{R}(\mathbf{x})$, computable from data alone, with

$$\mathbb{E}_\theta\big[\hat{R}(\mathbf{X})\big] \;=\; R(\theta, \delta).$$

Then risk comparisons become integrand comparisons.

This $\hat{R}$ is the **unbiased estimator of risk** (Berger $§$5.4.2). Its construction rests on Stein's Identity.

Stein's Identity (Berger $§$5.4.2)

Statement

Stein's Identity (univariate)

Let $Z \sim \mathcal{N}(\mu, 1)$, and let $g : \mathbb{R} \to \mathbb{R}$ be differentiable with $\mathbb{E}_\mu|g'(Z)| < \infty$ and suitable boundary conditions (specified below). Then

$$\mathbb{E}_\mu\big[(Z-\mu)\, g(Z)\big] \;=\; \mathbb{E}_\mu\big[g'(Z)\big].$$

The identity replaces a covariance with μ by a derivative. Heuristically: the density $\phi(z-\mu) = (2\pi)^{-1/2}\exp\{-(z-\mu)^2/2\}$ satisfies $(z-\mu)\phi(z-\mu) = -\phi'(z-\mu)$, so integration by parts swaps the $(z-\mu)$ factor for a derivative on g.

Proof (one-dimensional case)

Write the expectation as an integral and integrate by parts. Let $\phi(z) = (2\pi)^{-1/2} e^{-z^2/2}$.

$$\mathbb{E}_\mu[(Z-\mu)g(Z)] \;=\; \int_{-\infty}^{\infty} (z-\mu)\, g(z)\, \phi(z-\mu)\, dz.$$

Substitute $u = z - \mu$, so $z = u + \mu$:

$$= \int_{-\infty}^{\infty} u\, g(u+\mu)\, \phi(u)\, du.$$

The key fact: $u\phi(u) = -\phi'(u)$, since $\phi'(u) = -u\phi(u)$. So

$$= -\int_{-\infty}^{\infty} g(u+\mu)\, \phi'(u)\, du.$$

Proof, continued

Integrate by parts with $v = g(u+\mu)$ and $dw = \phi'(u)\,du$:

$$-\int g(u+\mu)\,\phi'(u)\,du \;=\; -\Big[g(u+\mu)\,\phi(u)\Big]_{-\infty}^{\infty} + \int g'(u+\mu)\,\phi(u)\,du.$$

Boundary conditions. The bracket vanishes if $g(u+\mu)\,\phi(u) \to 0$ as $u \to \pm\infty$. Sufficient condition: g has polynomial growth (since ϕ decays exponentially). For all the g we will encounter (linear, rational, smooth), this holds.

The remaining integral is $\mathbb{E}_\mu[g'(Z)]$. So

$$\mathbb{E}_\mu[(Z-\mu)\,g(Z)] \;=\; \mathbb{E}_\mu[g'(Z)]. \qquad \square$$

The multivariate version

Stein's Identity (p-dimensional)

Let $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$, and let $\mathbf{g} : \mathbb{R}^p \to \mathbb{R}^p$ have components g_i each differentiable with $\mathbb{E}_\theta|\partial g_i/\partial x_j| < \infty$ and suitable decay. Then for each coordinate i,

$$\mathbb{E}_\theta\big[(X_i - \theta_i)\, g_i(\mathbf{X})\big] \;=\; \mathbb{E}_\theta\!\left[\frac{\partial g_i}{\partial x_i}(\mathbf{X})\right].$$

Summing over i:

$$\mathbb{E}_\theta\big[(\mathbf{X} - \theta)^T \mathbf{g}(\mathbf{X})\big] \;=\; \mathbb{E}_\theta\big[\nabla \cdot \mathbf{g}(\mathbf{X})\big],$$

where $\nabla \cdot \mathbf{g} = \sum_{i=1}^p \partial g_i/\partial x_i$ is the divergence.

Proof. Fix i and the other coordinates. The conditional distribution of X_i given the others is $\mathcal{N}(\theta_i, 1)$. Apply the one-dimensional Stein identity to $z \mapsto g_i(\dots, z, \dots)$, then average over the remaining coordinates. $\square$

Building the unbiased estimator of risk

Setup

We restrict attention to estimators of the form

$$\delta(\mathbf{X}) \;=\; \mathbf{X} + \mathbf{g}(\mathbf{X}),$$

where $\mathbf{g} : \mathbb{R}^p \to \mathbb{R}^p$ is a "nice" correction term (differentiable, decay conditions for Stein's Identity). This is a natural parametrization: $\mathbf{g} = 0$ recovers δ^0, and any rule worth considering can be written this way.

For $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$:

$$\|\delta(\mathbf{X}) - \theta\|^2 \;=\; \|(\mathbf{X} - \theta) + \mathbf{g}(\mathbf{X})\|^2 \;=\; \|\mathbf{X} - \theta\|^2 + 2(\mathbf{X} - \theta)^T \mathbf{g}(\mathbf{X}) + \|\mathbf{g}(\mathbf{X})\|^2.$$

Taking expectations:

$$R(\theta, \delta) \;=\; p \;+\; 2\,\mathbb{E}_\theta\big[(\mathbf{X} - \theta)^T \mathbf{g}(\mathbf{X})\big] \;+\; \mathbb{E}_\theta\big[\|\mathbf{g}(\mathbf{X})\|^2\big].$$

The middle term has the θ-dependent cross product $(\mathbf{X} - \theta)^T \mathbf{g}$. Stein's Identity kills the θ dependence.

Applying Stein's Identity

By the multivariate Stein Identity:

$$\mathbb{E}_\theta\big[(\mathbf{X}-\theta)^T\mathbf{g}(\mathbf{X})\big] \;=\; \mathbb{E}_\theta\big[\nabla\cdot\mathbf{g}(\mathbf{X})\big].$$

Substituting into the risk formula:

$$R(\theta,\delta) \;=\; p \;+\; \mathbb{E}_\theta\big[2\nabla\cdot\mathbf{g}(\mathbf{X}) + \|\mathbf{g}(\mathbf{X})\|^2\big].$$

Stein's Unbiased Risk Estimate (SURE)

For $\delta(\mathbf{x}) = \mathbf{x} + \mathbf{g}(\mathbf{x})$ with $\mathbf{g}$ satisfying the conditions for Stein's Identity:

$$\widehat{R}(\mathbf{x}) \;=\; p \;+\; 2\nabla\cdot\mathbf{g}(\mathbf{x}) \;+\; \|\mathbf{g}(\mathbf{x})\|^2$$

is an **unbiased estimator of the risk**: $\mathbb{E}_\theta[\widehat{R}(\mathbf{X})] = R(\theta,\delta)$ for every θ.

What SURE accomplishes

The formula

$$R(\theta, \delta) \;=\; \mathbb{E}_\theta\big[\widehat{R}(\mathbf{X})\big] \;=\; \mathbb{E}_\theta\big[p + 2\nabla \cdot \mathbf{g}(\mathbf{X}) + \|\mathbf{g}(\mathbf{X})\|^2\big]$$

has remarkable properties:

(i) No θ on the right side. The function $\widehat{R}(\mathbf{x})$ depends only on $\mathbf{x}$. The expectation is taken under P_θ, but the *integrand* does not involve θ.

(ii) Pointwise dominance gives risk dominance. If $\widehat{R}_1(\mathbf{x}) \le \widehat{R}_2(\mathbf{x})$ for all $\mathbf{x}$, then $R(\theta, \delta_1) \le R(\theta, \delta_2)$ for all θ. The proof of dominance becomes a pointwise inequality on a function of $\mathbf{x}$.

(iii) SURE is computable from data alone. Given an observed $\mathbf{x}$, $\widehat{R}(\mathbf{x})$ is a number, no θ needed. This makes SURE useful in practice for *choosing* among estimators data-adaptively (e.g., model selection, denoising). We will return to this at the end of the lecture.

Verification of Stein's Identity for the James-Stein form

The right g

To compare δ^{JS} to δ^0, write $\delta^{JS}(\mathbf{x}) = \mathbf{x} + \mathbf{g}(\mathbf{x})$ where

$$\mathbf{g}(\mathbf{x}) \;=\; -\frac{p-2}{\|\mathbf{x}\|^2}\,\mathbf{x}.$$

So $g_i(\mathbf{x}) = -(p-2)\,x_i/\|\mathbf{x}\|^2$.

For Stein's Identity to apply, we need $\mathbf{g}$ to be differentiable with appropriate growth conditions. This $\mathbf{g}$ is smooth on $\mathbb{R}^p \setminus \{0\}$ but blows up at the origin. The singularity is integrable for $p \geq 3$ (since $\mathbb{E}_\theta[\|\mathbf{X}\|^{-2}] < \infty$ in this regime), and the boundary conditions are satisfied. We accept this technical point.

Computing the divergence

For each i:

$$\frac{\partial g_i}{\partial x_i} = -(p-2) \cdot \frac{\partial}{\partial x_i}\left[\frac{x_i}{\|\mathbf{x}\|^2}\right].$$

By the quotient rule:

$$\frac{\partial}{\partial x_i}\left[\frac{x_i}{\|\mathbf{x}\|^2}\right] = \frac{\|\mathbf{x}\|^2 \cdot 1 - x_i \cdot 2x_i}{\|\mathbf{x}\|^4} = \frac{\|\mathbf{x}\|^2 - 2x_i^2}{\|\mathbf{x}\|^4}.$$

Summing over i:

$$\nabla \cdot \mathbf{g}(\mathbf{x}) = -(p-2)\sum_{i=1}^{p} \frac{\|\mathbf{x}\|^2 - 2x_i^2}{\|\mathbf{x}\|^4} = -(p-2) \cdot \frac{p\|\mathbf{x}\|^2 - 2\|\mathbf{x}\|^2}{\|\mathbf{x}\|^4} = -\frac{(p-2)^2}{\|\mathbf{x}\|^2}.$$

SURE for the James-Stein estimator

Also:

$$\|\mathbf{g}(\mathbf{x})\|^2 \;=\; \frac{(p-2)^2}{\|\mathbf{x}\|^4}\,\|\mathbf{x}\|^2 \;=\; \frac{(p-2)^2}{\|\mathbf{x}\|^2}.$$

Combining into SURE:

$$\hat{R}^{JS}(\mathbf{x}) \;=\; p + 2\nabla\cdot\mathbf{g}(\mathbf{x}) + \|\mathbf{g}(\mathbf{x})\|^2 \;=\; p \;-\; \frac{2(p-2)^2}{\|\mathbf{x}\|^2} \;+\; \frac{(p-2)^2}{\|\mathbf{x}\|^2}.$$

Simplifying:

$$\boxed{\hat{R}^{JS}(\mathbf{x}) \;=\; p \;-\; \frac{(p-2)^2}{\|\mathbf{x}\|^2}.}$$

Taking expectations:

$$R(\theta,\delta^{JS}) \;=\; p \;-\; (p-2)^2\cdot\mathbb{E}_\theta\!\left[\frac{1}{\|\mathbf{X}\|^2}\right].$$

This is the formula we previewed in Lecture 5 and the start of today.

What we have established

The SURE machinery has reduced a risk computation to two steps:

(a) Compute SURE pointwise:

$$\hat{R}^{JS}(\mathbf{x}) \;=\; p - (p-2)^2/\|\mathbf{x}\|^2.$$

(b) Take expectation:

$$R(\theta, \delta^{JS}) \;=\; p - (p-2)^2\, \mathbb{E}_\theta\big[1/\|\mathbf{X}\|^2\big]\,.$$

For $p \geq 3$, the expectation in (b) is positive and finite (a noncentral chi-squared computation). Hence $R(\theta, \delta^{JS}) < p = R(\theta, \delta^0)$ for every θ.

We have **proved the Stein phenomenon** modulo the inverse-norm expectation, which we evaluate next lecture.

SURE in practice

Beyond proving theorems

Stein's unbiased risk estimate is more than a proof device. It is used in modern statistics for data-driven choice of estimators:

Tuning regularization. Given a family of estimators δ_λ indexed by a tuning parameter λ (e.g., ridge regression penalty, soft-thresholding level in wavelets), compute $\hat{R}_\lambda(\mathbf{x})$ and choose

$$\hat{\lambda}(\mathbf{x}) \;=\; \arg\min_{\lambda} \hat{R}_\lambda(\mathbf{x}).$$

Under mild conditions, $\hat{\lambda}$ approximates the oracle choice $\arg\min_\lambda R(\theta, \delta_\lambda)$.

Wavelet denoising. Donoho and Johnstone (1994, 1995) used SURE to choose thresholds in wavelet shrinkage estimators. The resulting **SureShrink** procedure achieves near-minimax denoising for a wide class of functions, and remains a standard reference in signal processing.

Empirical Bayes. SURE provides the loss-based criterion for choosing hyperparameters in hierarchical models. Donoho and Johnstone-style adaptive estimators are essentially empirical Bayes procedures justified by SURE.

Summary

What we built today

The setup. $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$, squared-error loss, $\theta \in \mathbb{R}^p$. The sample mean $\delta^0(\mathbf{x}) = \mathbf{x}$ is minimax for every p, but inadmissible for $p \geq 3$.

Stein's Identity. For $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ and smooth $\mathbf{g}$ with adequate decay:

$$\mathbb{E}_\theta\big[(\mathbf{X} - \theta)^T \mathbf{g}(\mathbf{X})\big] \;=\; \mathbb{E}_\theta\big[\nabla \cdot \mathbf{g}(\mathbf{X})\big].$$

Proved via integration by parts on the normal density. Eliminates explicit dependence on θ in risk calculations.

Stein's Unbiased Risk Estimate (SURE). For $\delta = \mathbf{x} + \mathbf{g}(\mathbf{x})$:

$$\hat{R}(\mathbf{x}) \;=\; p + 2\nabla \cdot \mathbf{g}(\mathbf{x}) + \|\mathbf{g}(\mathbf{x})\|^2.$$

Satisfies $\mathbb{E}_\theta[\hat{R}(\mathbf{X})] = R(\theta, \delta)$ for all θ. Risk comparisons reduce to pointwise comparisons of SURE.

James-Stein SURE computation. For $\mathbf{g}(\mathbf{x}) = -(p-2)\mathbf{x}/\|\mathbf{x}\|^2$:

$$\hat{R}^{JS}(\mathbf{x}) \;=\; p \;-\; \frac{(p-2)^2}{\|\mathbf{x}\|^2}.$$

Reading

Berger, Sections 5.4.1 (motivation, the Stein phenomenon) and 5.4.2 (the unbiased estimator of risk). The original references: Stein (1973) introduced the identity in modern form, and Stein (1981, *Annals of Statistics*) gave the comprehensive treatment.

Exercises: Berger $§$5.4 Exercises 52, 53, 54. For SURE in practice, Donoho and Johnstone (1995, *JASA*) “Adapting to Unknown Smoothness via Wavelet Shrinkage” remains the canonical reference.